State Methodology (10/19)

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This memo describes the state extension of the Budget Lab's ECEC model and its implementation on `state-inputs`, from survey data to equilibrium prices. Processing and calibration run nationally for the 2019 base year. State simulations cover children ages 0–4 in 50 states, DC, and New York City. New York City's separate run is excluded from national totals.

I compare against `main` at commit `3f54c9d` (18 August 2026) and describe `state-inputs` as implemented on 19 October 2026. Methods follow the national model except where described below.

The state extension separates national inputs, local targets, and model predictions:

- National estimates and assumptions: the care, employment, and price-wedge forests; behavioral parameters β and ρ; and supply parameters.
- Imposed for each geography: annual care hours in six care types, base wages, staffing rules, non-labor costs, and the employment path. Baseline employment follows group targets.
- Predicted by the model: the split between low- and high-price centers, equilibrium prices, equilibrium wages, policy-induced employment changes, and other responses to policy.

1 National foundation

The care model learns from the 2019 National Survey of Early Care and Education (NSECE), and the simulator's households come from the 2019 American Community Survey (ACS), to which I add state and county identifiers. Children receive one care label under the national model's provider-priority rules, which follow Borowsky et al. (2022); paid centers take priority over unpaid centers. Care outcomes distinguish seven non-parental types, each part-time or full-time, plus Parent Only.

`main` ranks families on nominal income. For state runs, I adjust income for local prices by dividing it by the 2019 BEA all-items regional price parity divided by 100, then apply national cutpoints built separately in NSECE and ACS. ACS households use their state's parity. NSECE identifies only the Census region, so NSECE households use the child-weighted mean parity of their region's states. New York City uses the New York metro parity, 114.9, and gets its own ranks and alpha.

National forests predict care conditional on caregiver employment, preserving joint care probabilities for the two youngest children. I combine these with employment probabilities to construct baseline choice probabilities, p_0. I retain the national utility specification and calibration targets: -0.15 for employment with respect to care cost and -0.05 for work hours with respect to income. Their empirical sources are undocumented. The price-adjusted calibration (interface `202610191203`) gives utility scale $\beta = 0.0003405$ and curvature $\rho = 0.0262$; both elasticities hit their targets. Alpha is a preference constant for each family and choice that reproduces p_0; state runs adjust it as described below. A family's care cost is its price wedge times the sector price times its care hours. The **price wedge** is the ratio of a family's payment to its sector's price, predicted from NSECE.

2 Center price tiers

`main` separates low- and high-price household center care at $4.75 an hour. For state runs, I classify posted prices and household payments using a threshold by region, urban density, and age band; prices at or above it are high-price. The tiers measure price, not observed quality. Regional rural and moderate-density samples are small, so I start with each region's high-density median and scale it by the national price ratio between density groups:

$$\underset{\text{(region, urban density, age band)}}{\text{threshold}} = \underset{\text{high urban density, age band}}{\text{regional median price,}} \times \frac{\underset{\text{this urban density, age band}}{\text{national median price,}}}{\underset{\text{high urban density, age band}}{\text{national median price,}}} \tag{1}$$

I calculate medians over positive prices, weighting centers by survey weight times full-time-equivalent enrollment. The posted price is the weekly rate divided by 40 hours, and a household's price is its weekly payment divided by its weekly hours. The age bands are 0–1, 2–3, and 4–5. Urban density is the NSECE community urban density (`COMM_URBAN_DENSITY`): high (85% or more of the nearby population is urban), moderate (30% to under 85%), or rural (under 30%). A center uses its own level, and a family uses its household's level. Regional cells contain 5–35 surveyed rural centers and 28–134 moderate-density centers per age band (unweighted). I assume the national ratio holds in every region. The care forest has no urban density feature.

3 Demand targets for each geography

A state's households carry national preferences, so I re-anchor each geography to its own level of care. The target is annual hours by care type. `main` has no demand targets.

3.1 Children in regular care

Children in regular care equal the population ages 0–4 times the NSCH rate times a five-hour factor. Population is the Census 2019 county estimate for ages 0–4. The National Survey of Children's Health (NSCH) rate is the share of children with at least 10 weekly hours of non-parental care, pooled over 2016–2019 within each state. New York City uses New York's rate because NSCH does not identify the city. Regular care in NSECE requires at least 5 hours, so I scale the rate by one national five-hour factor: NSECE children in regular care divided by NSECE children in care 10 or more hours a week. Both counts use NSECE child weights and cover children ages 0–4, excluding school programs. The factor is 1.098, based on 2,814 surveyed children in the numerator (unweighted). For the 50 states and DC, this implies 11.6 million children in regular care, or 59.2% of the estimated population of 19.6 million children ages 0–4.

3.2 Unpaid center children

Unpaid Center-Based care takes each geography's own program counts:

1. Pre-K is the larger of public-school pre-K from the Common Core of Data and an all-settings count. The all-settings count is Table 2 of the NIEER State of Preschool yearbook for a state, and the city's own count for New York City.
2. Funded Head Start and Early Head Start slots convert to enrolled children with the Office of Head Start enrollment ratios from the 2018–19 Program Information Report.

3. I remove home-based and family child care options, children age 5 and older, and Head Start children that pre-K already counts. I use one pooled Head Start overlap share: the slot-weighted mean across the 18 geographies with NIEER estimates.

I treat adjusted pre-K and Head Start enrollment as unpaid-center children. This can overstate that category: children who also use paid centers take the paid-center label, and some public programs charge families. The overlap adjustment removes duplication between pre-K and Head Start, not overlap with other care.

3.3 Other care types

I subtract unpaid-center children from total children in regular care, then allocate the remainder across the other five types using child-weighted p_0 shares within each geography. I use the same probabilities to calculate each type's part-time share. The mix reflects local household characteristics under the nationally estimated model.

3.4 Hours and re-anchoring

$$\text{annual hours of type} = \text{children in type} \times \big(\text{part-time share} \times 500 + \text{full-time share} \times 2{,}000\big) \quad (2)$$

Each geography's file has six rows: Center-Based, Unpaid Center-Based, Paid Home-Based, Other Unpaid, Other Paid, and Unpaid Home-Based. Low-price and high-price centers share one row.

At simulation start, I adjust alpha in two steps:

1. I change the preference constants at the geography's 2019 supply prices so that local prices alone do not change the initial choice probabilities, p_0.
2. I solve one constant δ for each care type, shared by every household in the geography, so that base-year hours match the file. Parent Only has $\delta = 0$. A household's alpha for a choice moves by the sum of δ over its children's care types. The two center tiers share one δ, so the model sets the split between them. In the same solve, I add one shift on working choices for each caregiver group (sex by four age bands), so each group's 2019 employment rate equals its employment-forest rate. The run stops if targets lack support or the adjustment fails to converge.

The joint solve targets care hours and employment rates; child counts and part-time shares remain model outcomes. In the earlier nominal-rank run `202610151655`, which matched care hours alone, mean absolute child-count deviations were 0.3–0.8% across care types, weighting the 52 geographies equally within each type. The largest absolute deviation was 9.2% (Other Paid in Vermont); part-time shares differed by at most 8.1 percentage points. These diagnostics do not validate the new price-adjusted joint solve.

The earlier run also explains the joint solve: matching care alone raised 2019 employment by 1.0–2.4 percentage points, averaging equally over 52 geographies within each caregiver group. Subsequent employment targeting reduced total non-parental care hours by 8.6% on average across those geographies at fixed 2026 prices (`baseline_consistency.R`). Matching both in 2019 aligns the care baseline with the employment rates from which yearly targets grow.

Initialization checks net-income reconstruction, recovery of p_0 after repricing, and consistency of base-year sector hours. Validation of the price-adjusted joint solve is pending; its 52-geography run ID and check file will be recorded here.

4 Supply and employment for each geography

Each sector uses fixed amounts of labor from workers with and without a bachelor's degree. Total demand across sectors determines each group's wage; paid-sector prices equal labor costs plus non-labor residuals. Unpaid centers charge households zero but contribute to labor demand; their financing is outside the model. I set each group's base-year labor supply equal to the labor required for re-anchored care hours. Each geography's supply file changes three national inputs: wages, staffing, and the residual.

4.1 Wages

I scale each group's national base wage by its state-to-national wage index ratio. Workers without and with a bachelor's degree have national base wages of $12.06 and $19.70 an hour. The wage index mixes May 2019 hourly medians from the BLS Occupational Employment Statistics for childcare workers, preschool teachers, and preschool and childcare administrators. The mix uses the national ACS share of each group's workers in each occupation. New York City uses the New York metro area.

4.2 Staffing

The staffing index is the state's required staff per child divided by the national average, with states weighted by children ages 0–4. The rules come from the 2017 Child Care Licensing Study (ICPSR 37700). For centers, required staff per child is the average of 1 over the child-to-staff ratio at five ages (11 months, 18 months, 35 months, 3 years, and 4 years), weighted by NSECE 2019 center enrollment at those ages. For homes, it is 1 over the maximum number of children in a family home. States without a family-home licensing ratio use an index of 1. New York City centers use the June 2019 city Health Code, and Idaho's center ratios come from its statutory points rule. I multiply national labor requirements per child-hour by the center index for all three center sectors and by the home index for paid homes, applying each index equally to both worker groups. Classrooms have more staff than the limits require. I use the licensing limits because they are set by state rule and available for every state.

4.3 Non-labor costs

`main` uses one national residual per sector: the hourly price minus labor cost per child-hour. These are $0.51 for low-price centers, $4.61 for high-price centers, and $2.83 for paid homes. I scale each sector's national non-labor residual by the geography's 2019 BEA housing-services price parity divided by 100. New York City uses the metro value, and unpaid centers keep a zero residual. The factor runs from 0.55 to 1.70 across the 52 geographies. The residual covers non-labor costs and provider margins. I use housing-services prices as a proxy for geographic variation in non-labor costs.

Across 46 states, housing scaling raises the unweighted Pearson correlation with National Database of Childcare Prices (NDCP) prices from 0.83 to 0.89 and lowers weekly root mean squared error from $41.6 to $34.1. However, the ratio of child-weighted mean model to NDCP prices rises from 1.09 to 1.12. The comparison uses base-year posted prices, fixed model enrollment weights, and child-weighted county medians, including imputed prices (`state_data/code/16_ndcp_model_prices.R`). Age bands and full-time definitions align only approximately.

4.4 Employment path

A geography's projected employment rate, by sex and age band, is the national projected rate times the state-to-national ACS rate ratio. The national series uses Current Population Survey history, projected forward using CBO's long-term labor force participation projections. The ratios use ACS two-year pools and remain at their 2024 values thereafter. I aggregate the resulting path to four caregiver age groups by sex, then apply growth since 2019 to each group's base-year employment rate, the weighted mean of its employment-forest probabilities.

5 Simulation

Each year, I update household weights, earnings, income, and prices using national growth factors. Each geography is its own market, with no movement of workers or families between geographies. This excludes cross-border labor-market links, especially relevant for DC. The solver starts from the prior year's prices. Group-specific utility shifts on working choices match the baseline employment targets and stay fixed across policy scenarios. Baseline and policy runs use the same Gumbel shocks. Supply links demand to price through labor:

$$\text{wage of group} = \text{base wage} \times \text{wage growth} \times \left(\frac{\text{labor demand}}{\text{base-year labor}} \right)^{1/\text{supply elasticity}} \tag{3}$$

The supply elasticities are 4 for workers without a degree and 2 for workers with one, assumed values from Borowsky et al. (2022). I keep them national in every geography. The residual grows with the CPI.

6 Open questions

1. Should state base-year targets sum to national care hours, or should I report the discrepancy?
2. Should local prices change baseline care probabilities, or should local alpha absorb their effect before demand targeting?
3. Is regional price adjustment in NSECE close enough to state adjustment in ACS? The forests learn from regional parities but apply state parities.